

Video Transcripts

Week 6: Agentic AI Fundamentals

Video 1: Why Agentic AI, Why Now

Hello learners and welcome to this week. In this session, we begin with the fundamental shift from traditional AI systems to Agentic AI systems. To understand Agentic AI, we must first recognise that AI has evolved in three distinct layers.

The first layer is classical machine learning. These systems are designed primarily for prediction. They analyse historical data and generate outputs such as forecasts or classifications. For example, in banking, machine learning models predict the probability of loan default. However, these systems do not act on their predictions.

The second layer is generative AI. These systems introduce the ability to create content such as text, images, code, and analysis. However, they remain fundamentally passive. They respond to prompts but do not execute workflows or take actions.

The third layer is Agentic AI, which introduces a critical capability: autonomous execution. At the core of Agentic AI is the planning component, which allows systems to break down goals into structured steps. Unlike generative AI, which responds in a single step, agentic systems can decompose tasks such as 'resolve a customer complaint' into retrieval, analysis, resolution, and system update.

Alongside planning, another key component is tool usage. Agents are no longer limited to internal knowledge. They can interact with external systems such as APIs, databases, and enterprise applications. For example, a support agent can check order status directly from a CRM system instead of generating a suggestion.

This combination of planning and tool usage fundamentally changes the role of AI. It moves from assistance to execution.

A useful way to understand this shift is through an analogy. Generative AI is like a skilled analyst producing outputs on demand. An AI agent is like a specialist that completes a defined task independently. Agentic AI, however, is like a coordinated team that executes an entire business process end to end.

For example, in customer service, generative AI drafts replies. An AI agent resolves a single query. But an agentic system handles the full workflow – interpreting the issue, retrieving data, executing resolution, and updating systems automatically.

This is possible because modern systems now integrate planning models with tool ecosystems and real-time enterprise data.

2026 marks an inflection point because these components – planning, tool integration, and system interoperability – have matured enough for production-scale deployment.

The key shift is this: work is moving from human execution to machine-driven execution.

In summary, agentic AI marks a shift from prediction and content generation to autonomous execution. Unlike earlier AI, it combines planning and tool usage to break down goals, interact with systems, and complete workflows independently. This moves AI from assisting humans to executing tasks, enabled by advances in integration and real-time data.

Video 2: AI Agents vs Agentic AI – Core Distinction

Hello learners. At this stage, it is essential to understand a core distinction: the difference between AI agents and Agentic AI systems. While the terms are often used interchangeably, they operate at very different levels of capability and structure.

Let us begin with AI agents. An AI agent is a system that autonomously performs tasks by designing its own workflow and using available tools. At its core, it is typically a large language model combined with goals, tools, and memory, enabling it to reason and act without step-by-step human instruction. This is what differentiates an agent from a traditional model. It is not just generating responses; it is making decisions, solving problems, interacting with external systems, and executing actions in real environments.

An AI agent operates through a set of core capabilities. It takes a goal, plans how to achieve it, uses tools to interact with systems, and applies reasoning to adapt as needed. Among these, tool usage is particularly important because it allows the agent to move beyond internal knowledge and perform real actions.

For example, a travel booking agent does not just suggest flights. It can search across platforms, compare prices, and complete bookings by interacting with external APIs. This ability to act is what makes agents operational rather than informational.

AI agents can also be understood through a practical taxonomy based on how they make decisions. The simplest type is the reflex agent. This type reacts directly to inputs using predefined rules. It is fast but highly inflexible, as it cannot adapt beyond its programmed responses. A basic chatbot that replies based on keywords is a typical example.

The next level is the model-based reflex agent. This type maintains an internal representation of its environment, allowing it to handle situations where not all information is immediately visible. For instance, a warehouse management system that tracks inventory changes over time uses this approach to maintain continuity.

More advanced are goal-based agents. These agents evaluate different possible actions and choose the one that best achieves a specified objective. This makes them far more flexible and capable of handling complex tasks. Most enterprise applications today fall into this category because they require structured decision-making rather than simple reactions.

At the most advanced level are utility-based agents. These agents go beyond achieving a single goal and instead optimise across multiple competing objectives, such as cost, speed, and efficiency. For example, in pricing or logistics, an agent may need to balance profitability with delivery time and resource constraints. This makes utility-based agents particularly valuable in scenarios where trade-offs must be managed dynamically.

From a business perspective, most enterprise implementations begin with goal-based agents because they are easier to define and control. Utility-based agents are introduced when organisations are ready to handle more complex optimisation problems.

Now, let us shift to Agentic AI. Agentic AI operates at a higher level. It is not a single agent performing a task. Instead, it is a system where multiple specialised agents are orchestrated to achieve a broader objective. The key concept here is coordination. Rather than one agent doing everything, multiple agents collaborate, each focusing on a specific role.

In an agentic system, several types of agents typically work together. Orchestrator agents are responsible for planning and delegation. They break down a large goal into smaller tasks and assign them to other agents. Executor agents carry out those tasks, interacting with tools and systems as needed. Validator agents check outputs to ensure accuracy, compliance, and consistency. Communication agents handle interactions with users or external systems, ensuring that information flows smoothly across the workflow.

The practical difference between an AI agent and Agentic AI becomes clear when we look at real examples. An individual AI agent might handle a single task, such as booking a meeting. It takes a request, checks availability, and schedules the meeting. In contrast, an agentic system can manage an entire workflow. In a recruitment scenario, instead of just scheduling interviews, the system can screen resumes, shortlist candidates, coordinate interview scheduling, and generate hiring recommendations. This is not a single action but an end-to-end process managed through orchestration.

Another important element in agentic systems is validation. Because multiple agents are involved, outputs from one agent often become inputs for another. Without proper validation, errors can propagate across the system. Validator agents play a critical role in ensuring that each step meets the required standards before the workflow moves forward. This adds a layer of control that is essential for enterprise use.

This shift from single-agent execution to multi-agent orchestration also introduces new governance considerations. In a single-agent setup, errors are usually contained within a specific task. In a multi-agent system, however, errors can spread across the workflow and amplify if not detected early. This makes observability and auditability essential. Organisations need to be able to track what each agent did, how decisions were made, and where issues occurred. Clear accountability must also be established, with defined ownership for outcomes.

So the distinction can be summarised simply. An AI agent executes a task using planning, reasoning, tools, and memory. Agentic AI orchestrates multiple agents, coordinating their actions and validating their outputs to achieve a larger goal. One focuses on task execution, while the other focuses on workflow transformation.

The key insight is that Agentic AI represents a shift from isolated intelligence to coordinated intelligence. It allows organisations to move from automating individual tasks to redesigning entire processes. However, with this increased capability comes increased complexity, particularly in governance and control.

In summary, AI agents are autonomous systems that use reasoning, planning, tools, and memory to perform tasks, while Agentic AI orchestrates multiple specialised agents to manage end-to-end workflows. This distinction shapes how organisations design and deploy AI at scale.

Video 3: How Agents Work: Perception and Tool Use

Hello learners. To understand Agentic AI at a fundamental level, we need to move beyond definitions and look at how an AI agent actually works internally.

An AI agent is not just a language model generating responses. It is an execution system that continuously interacts with its environment, makes decisions, takes action, and improves over time.

At the core of this system is a continuous loop: observe, reason, plan, act, and reflect. Every capability we discuss is built on this loop.

Let us begin with the first component, perception. Perception is how the agent takes in inputs from its environment and converts them into usable context. These inputs can include natural language queries, structured enterprise data, documents such as PDFs or contracts, API responses, and even real-time streaming data.

For example, in a banking system, perception involves reading transaction records, account history, fraud signals, and customer profiles. In a customer support environment, it includes chat messages, ticket history, and product information. The key idea is that perception transforms raw data into context that the agent can reason about. Without perception, the agent has no awareness of what is happening around it.

The next component is planning. Planning is where the agent breaks down a complex goal into smaller, ordered subtasks. Not every task requires explicit planning. For simple queries, agents may rely on iterative reasoning instead. But for complex workflows, planning becomes essential.

For instance, if the goal is to generate a financial report, the agent may first plan to retrieve data, then analyse it, and finally generate the report. Planning provides structure and direction, especially in multi-step enterprise workflows.

Closely connected to planning is reasoning. Reasoning determines how the agent decides what to do next and how it adapts as new information becomes available. Two important reasoning patterns are commonly used. The first is ReAct, which stands for Reason and Act.

In this approach, the agent alternates between thinking and acting. It analyses a situation, takes an action such as calling an API, observes the result, and then decides the next step. This makes it highly effective in dynamic environments.

The second approach is ReWOO, or Reasoning Without Observation in advance. Here, the agent creates a complete plan upfront and then executes it step by step. This is better suited for structured workflows where the sequence of steps is predictable. The choice between these approaches depends on the nature of the task.

The fourth component is tool calling, which enables the agent to interact with external systems. This is what moves an AI system from simply generating responses to actually performing actions. Tools can include web search engines, APIs, databases, CRM platforms, ERP systems, and even other agents.

For example, in an e-commerce scenario, an agent might check product availability through an inventory API, validate pricing through another system, and confirm a transaction using a payment gateway. In enterprise environments, tool calling often maps directly to existing system integrations. It allows agents to operate within real workflows rather than remaining isolated from them.

The fifth component is memory. Memory allows the agent to store and retrieve past interactions, enabling continuity across tasks and sessions. There are typically two forms of memory. Short-term memory maintains context within a single interaction, ensuring that conversations remain coherent. Long-term memory stores historical data, such as previous interactions, preferences, and organisational knowledge.

For example, in customer support, memory allows the agent to recall past issues and avoid repeating the same troubleshooting steps. In an enterprise context, memory effectively becomes institutional knowledge embedded within the system. Technically, this is often implemented through databases or vector-based retrieval systems, allowing the agent to recall relevant information when needed.

The final component is learning. Learning allows the agent to improve its performance over time. This happens through feedback from users, evaluation of outcomes, and exposure to new data.

For example, in fraud detection, an agent can learn from confirmed fraud cases and adjust its detection logic accordingly. In enterprise workflows, this creates a compounding advantage. The more the system operates, the more data it generates, and the better it becomes at making decisions. Learning transforms the agent from a static system into one that evolves continuously.

When we combine these six components, we get a complete execution system. Perception provides context, planning structures the task, reasoning drives decision-making, tool calling enables real-world action, memory ensures continuity, and learning improves performance over time. This is fundamentally different from a traditional language model. An AI agent is not just answering questions. It is operating within workflows, interacting with systems, and producing outcomes.

It is also useful to map these components to an organisational context. Tool calling corresponds to your existing system integrations, such as APIs and enterprise platforms. Memory represents institutional knowledge, including historical data and accumulated insights. Learning reflects

continuous process improvement, where systems become more efficient and accurate over time. This mapping helps ground agentic AI in practical enterprise terms.

Now let us look at how these agents are structured at a system level. This brings us to agentic architecture, which represents a shift in how systems are designed. Traditionally, organisations followed a data-to-model-to-product approach. They built models first and then tried to find use cases. In contrast, the agentic approach starts with the desired business outcome. From there, we design the agent, identify the required data, and then select or build the appropriate models. This reverses the development process and aligns it more closely with real business needs.

Most enterprise systems adopt a hierarchical architecture. At the top, an orchestrator agent manages the workflow. It breaks down tasks into smaller components and assigns them to specialised agents. Executor agents perform specific actions such as retrieving data or processing information.

Validator agents check outputs for accuracy, compliance, and consistency.

The results are then aggregated and returned through the system. This structure ensures both autonomy and control, allowing complex workflows to be managed effectively.

Unlike traditional systems that follow rigid, linear flows, agentic workflows are adaptive and event-driven. Agents can adjust their plans in real time based on new information. They can re-plan mid-execution if conditions change. This flexibility is one of the key advantages of agentic systems, especially in dynamic enterprise environments.

However, it also introduces new considerations. Because agents can take different paths to reach a goal, governance becomes critical. Human-in-the-loop checkpoints must be deliberately designed into workflows to ensure oversight, especially in high-risk scenarios.

The key insight is that agents are not single models. They are systems composed of multiple capabilities working together in a continuous loop. This is what enables them to handle real-world complexity and operate at enterprise scale.

In summary, AI agents are built on six core components: perception, planning, reasoning, tool calling, memory, and learning. Together, within a structured architecture, they form execution systems capable of interacting with real environments, making decisions, and continuously improving. Understanding these fundamentals is essential before moving into more advanced topics such as multi-agent systems and enterprise deployment.

Video 4: Multi-Agent Systems and Agentic RAG

Hello learners. As systems scale, a single agent is often not enough to handle real-world complexity. This is where multi-agent systems come in. A multi-agent system is a collection of specialised agents that collaborate to complete tasks that no single agent could reliably handle alone. Instead of relying on one general-purpose agent, we design multiple focused agents, each responsible for a specific role, and coordinate them to work together.

At the centre of this system is orchestration. Orchestration is what manages how tasks are broken down, assigned, and coordinated across different agents. Without orchestration, multiple agents would simply operate in isolation. With orchestration, they function as a coherent system.

In most enterprise setups, you will see four key roles emerge. Orchestrator agents are responsible for planning and delegation. They decide what needs to be done and which agent should handle each part. Executor agents carry out specific tasks using tools and APIs, such as retrieving data or updating systems. Validator agents check outputs to ensure accuracy, consistency, and compliance. Communication agents act as interfaces, interacting with users or external systems and passing information back into the workflow.

As these systems grow, communication between agents becomes essential. In 2026, interoperability is enabled through protocols such as ACP, Agent-to-Agent communication (A2A), and the Model Context Protocol (MCP). These allow agents built on different frameworks to exchange information and coordinate actions seamlessly, which is critical in enterprise environments with diverse systems.

There are also different architectural approaches. In simpler scenarios, a flat architecture allows multiple agents to operate in parallel on independent tasks. However, for more complex workflows, hierarchical architectures are more effective. A central orchestrator manages multiple layers of agents, enabling better coordination, control, and scalability. This is particularly useful for enterprise workflows that involve dependencies and multi-step decision-making.

To make this concrete, consider a contract management system. One agent retrieves relevant clauses from documents using search tools. Another analyses those clauses to identify potential risks. A third drafts suggested revisions based on policies and context. A validator agent checks the output for compliance and accuracy. The orchestrator coordinates all agents, ensuring the workflow progresses correctly and outputs are combined into a final result. This distributed approach is far more reliable than relying on a single agent.

Building on this, Agentic RAG extends how agents interact with information. Traditional retrieval-augmented generation follows a simple pattern: retrieve a relevant document and generate a response. This is a single-step, single-source process, which limits its ability to handle complex queries.

Agentic RAG introduces multi-step, agent-driven retrieval. The agent decides what to retrieve, from which sources, and in what sequence. This makes the process dynamic and adaptive. It can interact with multiple tools, such as search APIs, databases, and internal systems, to build a more complete understanding before generating a response.

A key capability here is adaptive planning. The agent continuously refines its retrieval strategy based on what it finds. For example, in legal research, it may start with case law, then move to regulatory documents, and finally consult internal policies. Each step builds on the previous one to improve context and accuracy.

Another important feature is corrective retrieval. The agent evaluates the quality of retrieved information and, if needed, re-queries to improve results. This self-correction loop helps handle incomplete or ambiguous queries more effectively.

From an enterprise perspective, Agentic RAG enables systems to handle queries that span multiple repositories and knowledge bases. It moves beyond single-source answers and supports more complex, real-world information needs.

However, this added capability increases the risk of errors or hallucinations. As retrieval becomes more complex, validation becomes essential. Outputs must be checked through built-in validation steps rather than assumed to be correct.

The key takeaway is that multi-agent systems and Agentic RAG represent a shift from isolated intelligence to coordinated intelligence. Rather than relying on a single model to do everything, we distribute responsibilities across specialised agents and allow them to collaborate. This not only improves performance and reliability but also enables systems to handle far more complex, real-world tasks. As enterprise use cases grow in scale and complexity, this combination of multi-agent orchestration and intelligent retrieval becomes essential for building systems that are both powerful and practical.

Video 5: Synthesis and Action Item

Hello learners. We have now reached the synthesis stage of this module, where we bring together all the concepts we have explored and connect them into a unified understanding of Agentic AI.

Let us begin by revisiting the evolution of artificial intelligence. We started with classical machine learning systems that focus on prediction. These systems analyse historical data and generate forecasts or classifications. They are powerful, but they do not take action.

We then moved to generative AI systems, which introduced the ability to create content such as text, images, code, and analysis. However, these systems still depend on human prompts and do not execute tasks independently. Now, we are in the era of agentic AI. This is where systems move beyond generation and begin to execute actions. They can plan, decide, and operate across multiple steps to complete workflows. The key shift here is important. Work is no longer just supported by AI. It is increasingly being executed by AI systems.

Next, we examined the difference between AI agents and agentic AI. An AI agent is a single autonomous execution unit. It receives a goal and determines the steps required to achieve it using reasoning, tools, and memory.

Agentic AI, on the other hand, is a coordinated system of multiple agents working together. Instead of one system performing one task, we now have multiple specialised agents collaborating to complete entire workflows. This introduces a new concept: orchestration. One agent may plan, another may execute, and another may validate outcomes.

We then explored how individual agents work internally. Each agent operates using six core components: perception, reasoning, planning, tool calling, memory, and learning.

Perception allows the agent to understand inputs such as data, documents, or API responses. Reasoning helps the agent interpret information and decide what to do next. Planning breaks complex

goals into structured steps. Tool calling allows interaction with external systems such as APIs and databases. Memory enables continuity across tasks and interactions. Learning allows continuous improvement over time based on feedback.

Together, these components operate in a continuous loop of observe, reason, plan, act, and reflect. This loop is what enables agents to move from simple responses to meaningful execution.

We also studied multi-agent systems and Agentic RAG. Multi-agent systems involve different agents performing specialised roles. For example, orchestrator agents manage workflows, executor agents perform tasks, validator agents check outputs, and communication agents interact with users and systems.

Agentic RAG extends retrieval systems by allowing agents to decide what information to retrieve, from where, and in what sequence. Unlike traditional retrieval systems, this approach is multi-step and adaptive. Together, these systems enable distributed intelligence across enterprise environments.

We then examined enterprise use cases. Across industries such as finance, HR, supply chain, procurement, and customer service, agentic AI is transforming how work is done. The key insight is not task automation, but workflow compression. Entire processes that once required multiple teams and systems can now be executed in coordinated steps by agentic systems.

Finally, we introduced a structured way to move from theory to application. To implement agentic AI effectively, organisations must identify real processes and evaluate them using structured frameworks such as readiness assessment and governance evaluation. This ensures that adoption is not experimental, but targeted and controlled.

Across everything we have studied, four governance principles remain critical. First is accountability. Every agent must have a clearly defined owner responsible for its outputs. Second is observability. Agent behaviour must be monitored and traceable in real time. Third is human-in-the-loop design. Critical decisions must include human oversight. Fourth is data integrity. The quality of agent decisions depends entirely on the quality of data and tools they access.

To conclude, the key takeaway is this. Agentic AI is not just a technological advancement. It represents a shift in how work is structured and executed inside organisations.

The next step is application. You are now expected to identify real processes in your environment, evaluate their readiness, and begin thinking about how agentic systems could be applied responsibly. Ultimately, the question is no longer whether agentic AI will be adopted. The real question is how effectively it will be designed, governed, and integrated into enterprise systems.

Video 6: Future Outlook on AI Agents

Hello learners, welcome to this session. In this session, we will talk about the future outlook of AI agents.

As we have seen in the previous three demos, we have gone through Gumloop, Langflow, N8n, Make.com, and GFile. These are the five platforms that I have chosen to demonstrate to you. But these are not the only platforms available this year. Every week, a new platform is coming with new capabilities, with new offerings, with different pricing, with different inference, and with different use cases.

Every platform is trying to win over agentic AI workflows. So I would list a few more platforms so that you can go and check whether they are useful for you and your businesses and so on. One of them is Zapier Agents, then Lindley, Relevance AI, MindStudio, RelayApp, DeFi, Flowsy, Active Pieces, Microsoft Copilot, Lyr and SigmaMind.

These are different platforms which are doing the same thing. As in the previous five platforms that we have seen, they are helping you to create AI-augmented workflows. And if you see, as a whole, what agentic AI is changing in the business, the workflows were already there in businesses earlier.

Those are deterministic, logical flows that happen in businesses and so on. So earlier, let us say one trigger has come that your inventory is down and you need to order it. Earlier, we needed to have deterministic triggers – order is down, then the human will come and then order for replenishment.

By the invention of generative AI, we have created an LLM, a large language model. If you see a large language model as a whole, it acts as a translator to the machine, so that now the machine understands natural language. Now this large language model can be appended into the workflows, with the ability to understand human input, the ability to understand unstructured input, the ability to interpret input, reason it, and also we can attach a tool to act on it.

So your existing business workflows can now incorporate this large language model. Sometimes large language models are not needed, so we use a small language model. Basically, these language models understand natural language or unstructured language and, on your behalf, act on the next workflow. That is their placement in the workflow.

Making this LLM into an AI agent is the understanding that is common in all five toolsets that we have demonstrated. And maybe six months down the line, new tools will come and new tools will become famous. But the underlying understanding of this workflow and where LLMs sit in this place will remain the same.

As an example, as of now, while I am recording this video, the most popular and the most used LLM platform is Anthropic Claude. But one year before, it was ChatGPT by OpenAI. We do not know who will be the number one company serving best in this space one year later, six months later, or two years later.

But the concept of the large language model has remained constant from 2019 – 2017 onwards. What these new companies are doing is doing it in a better way. They are doing it in a cost-effective way. They are improving how useful it is becoming to businesses and individuals. Similarly, this is how agentic AI is being used.

So I will give you an analogy. If an LLM is a natural language understanding program, then you can understand this as a brain. Your LLM is a brain. And now your brain can instruct your different limbs and your different senses to receive the input and act on it.

On this brain, we can build or attach our existing toolsets, limbs, or perceiving agents. So a user input, or a trigger from a notification, or a trigger from a Google Sheet, Microsoft Excel, or anything that is being added into the existing enterprise data flow, data management, or enterprise management system can be considered as a perceiving tool or an action tool.

Using this additional brain feature, we are trying to create a new body for a computer, which now has a much more evolved brain closer to human intelligence. This was not heard of earlier before this decade, because all of the workflows, automation, or deterministic systems were deterministic in nature.

Due to this de-bottlenecking using large language models, and the plethora of models now available, the impact on businesses will be profound and the whole design thinking process will change. Your determinism and the given bottleneck of intelligence are now being slowly removed.

Later on, two years or three years down the line, it will be very easy to deal with these things. But the concepts of the brain, the limbs, and the toolsets will remain the same. We can have more complications. You can have one agent talking to another agent. You can have several agents talking to another agent.

You have a discussion mechanism that is already going on in experimental settings nowadays also. Later on, it will evolve in this manner. But if we understand their core functioning, we will be able to grasp all the new technologies in the future of agentic AI and agentic workflows in a much easier manner.

If we have clarity on how LLMs are integrating into business workflows, then we can understand this transformation more effectively. Precisely, I cannot give you a very specific example for your business, but you need to understand this. You need to inculcate this thinking into yourself – how your workflow can be automated and how the use of a natural language understanding computer model can be incorporated.

Once you have that idea in your mind, you can go back to these toolsets and sandbox it. It will require some time. You will hit some roadblocks and bottlenecks. The bigger LLM model that you have already subscribed to, or that you are using on a free plan, can help you in de-blocking, de-bottlenecking, and clearing your errors.

Once that sandbox idea is ready, then you can think about deploying it at scale. The key takeaway here is to understand the placement of the large language model and understand the use of generative AI in the business ecosystem, so that your whole thinking paradigm will change. It is unprecedented and unlike previous decades so far, because human and machine interaction happened only through code.

I hope I am able to summarise and convey this change in thinking and design thinking patterns in creating workflows. It will unlock a lot of business potential. Many companies are replacing their employees with AI agents now. Some of them are failing, some of them are succeeding.

If you can create your own agents, test them, and deploy them effectively, you might unlock greater business value and even a new business domain in itself.

Video 7: Final Thoughts

Hello learners.

We have now reached the conclusion of this module on Fundamentals of Agentic AI. To close, let us connect the key ideas.

We began by understanding the evolution of artificial intelligence. AI has progressed from predictive systems, to generative systems, and now to agentic systems that can take action. The critical shift we explored is this: AI is no longer only producing outputs, but increasingly executing workflows.

We then studied AI agents as individual execution units. These agents operate using core components such as perception, reasoning, planning, tool calling, memory, and learning. Each component plays a specific role. Perception enables understanding of inputs. Reasoning enables decision-making. Planning structures multi-step tasks. Tool calling connects agents to external systems like APIs and databases. Memory ensures continuity across interactions. Learning enables continuous improvement over time.

Together, these components form a continuous execution loop of observe, reason, plan, act, and reflect.

We then moved to agentic AI systems, where multiple agents work together in coordinated structures. Here, orchestration becomes critical, with different agents handling execution, validation, communication, and control.

We also explored multi-agent systems and Agentic RAG, which enable distributed intelligence and multi-step retrieval across enterprise systems. Across all enterprise applications – from finance and HR to supply chain and customer service – the consistent pattern is workflow transformation. Not just automation of tasks, but redesign of how entire processes operate.

However, capability alone is not enough. Governance is essential. This includes accountability for agent decisions, observability through monitoring systems, human-in-the-loop oversight for critical actions, and strong data integrity. Without these, scalability becomes risk.

To conclude, agentic AI represents a fundamental shift in how enterprises operate. It is not just a new technology layer, but a new operating model for execution.

The next step is application. The focus now moves to identifying real processes, evaluating readiness, and designing systems responsibly. Ultimately, the key question is not whether agentic AI will be adopted, but how effectively it will be shaped within your organisation.

Video 8: Gumloop

Hello learners, welcome to this session. In this session, we will take a demonstration on agentic AI platforms. Gumloop is an example that I will be using in this session, where we will be displaying perceive, reason, and action, and the use of several LLMs in order to make your own agents.

Let us now move towards the demo part. So we will open Google and write down Gumloop. We will open this Gumloop.

You need to log in using your email ID. I have already logged in. Therefore, it is welcoming me back. I have also made an agent named Flight Booking. First, I will demonstrate this agent. Then we will go back and start from scratch on how to make your own agent.

So my agent looks like this. You can see there are three columns in this panel. The first column is about new chat, file generated, inbox, and so on. The second panel is like a chatbot. And the third panel is the agent that I have built.

As you can see in the third panel, we have different kinds of LLMs that are available. The selected LLM at this point in time is Claude 4.6. You are not required to select this. You have several different LLMs available on this platform, including Claude, Anthropic, OpenAI, Google, Qwen, and so on.

You can choose any of these LLM providers and any of the LLMs among these providers. So by default, let it be Claude 4.6. This is the latest one.

Next, in this box, I have given a system prompt describing what this bot will do. I am saying that you are a helpful travel agent. Perceive the user request. Reason the best option. Use Firecrawl to search and find flights. Summarise the cheapest option and learn from user experience.

If you notice, the agent concept is first perceive or sense from the user or any other input, then reason the best option, and then use the tool. So Firecrawl is used as a search tool similar to Google. There are different search tools available apart from Google Search, Bing Search, Firecrawl, DuckDuckGo, and so on.

Then it will search flights and summarise the cheapest option. And the last one is very important – learn from the user preferences. I am allowing self-updates. There are triggers you can add and manage. You can add more apps.

I will show you that there are 93 apps available here. Your input can be from your Gmail. The input can be from your Google Sheet. The input can be from your Google Drive, Google Calendar, and so on. So instead of you providing the input, the input might already be sitting in these apps.

I will run through the different kinds of apps available, from Outlook to Reddit to YouTube to Jira to Slack. Say, for example, your conversation is going on in Slack and you want to summarise it and act on it. You can use your Slack conversation as an input and provide it to the LLM to reason and respond on your behalf.

As of now, I made a very simple agent to search the best flight for me. You can see there are other abilities. Web search is on. Image generation is off because there is no need to generate images in a flight search. Fast conversation is on.

I have run this, and I will show you the output. The history of my chat is here. As you can see, I have asked the agent to book a cheap flight to Mumbai on May 25 from Indore. As you can see, it has called 12 tools, exchanged two messages, summarised the cheapest option, provided the key insights, provided me the links where I can go and book my flight, and also provided recommendations.

You can add more capabilities to it. For example, you can add your agency to book the flight directly by adding your credentials and so on.

Now I will go back and start a new agent. So now we will build our own agent, as I built my Flight Booking agent earlier. To build my own agent, I will go to this agent section and create a new agent.

Once I create this new agent, I have to select the LLM provider. So I can select Gemini Flash and I have to add my instruction. For example, in this case, I am creating an agent for B2B lead qualification.

So I will add a system prompt. I have already made a system prompt. It says: You are an intelligent B2B lead qualification agent. Always follow perceive, reason, act, and learn.

What do you perceive? Understand the user's lead and company name. Plan what information is needed to qualify the lead – industry name, company size, decision maker, pain points, budget, and so on.

Act: Use Firecrawl or a web search tool to research the company website, LinkedIn, news, and recent updates. Remember the lead details, user preferences, target industry, deal size, and so on.

Then score the lead from 1 to 10 based on fit. Give a clear summary, company overview, key decision maker, potential pain points, next recommended action, and suggest a personalised outreach message if the score is 7+. Always be professional and data-driven and start by asking for the company name or website for research.

I have added this system prompt and allowed self-updates. I will add an app. I have added Firecrawl, and AI skills editing is on.

You can create with AI, upload files, write skill instructions, add skills, or use existing skills if you have already done so in Gumloop. Sub-agents are not needed. The abilities are all on. I will make image generation off and keep past conversation on, and so on.

I will save this agent later on. You can rename this agent. Now we will test this agent by asking it to research a company. Let us say, just for the sake of testing, 'Research Acme Corporation'.

It starts this action. You can see tools are called, messages are generated, and intermediate reasoning is taking place. Whatever system prompt we have given, it is trying to provide a summary accordingly.

You can further customise this agent and add more tools. You can add more actions to it depending on your use case. This kind of loop of perceive, reason, act, and learn can be easily experimented with here, using your specific use case and the customisation available.

Your agent will be listed here. You can rename it as a B2B Research Agent. You can use these agents within your workflows later on to automate processes using LLMs.

In conclusion, in this demo we have seen an agent, how we customise the agent, how we use tools, and how we deploy the agent on the web system called Gumloop.

Thank you.

Video 9: Langflow

Here is the transcript with punctuation, capitalisation, spacing, and obvious transcription errors corrected, while preserving the original wording and intent as closely as possible. I have also grouped related content into logical paragraphs and avoided one-line paragraphs.

Hello learners, welcome to this session. In this session, we will talk about another AI agentic demonstration. As in the previous session, we have spoken about Gumloop.

Here, we will demonstrate another agent called Langflow. The core difference between the agentic platform that we demonstrated previously and this one is that here you have the ability to choose your models and run them offline on your laptop, your system, an industrial system, or an enterprise system. This gives you much more flexibility. Your data security is way better than sending your data to online servers.

So let me directly jump to the demo and show you how it works. I will search for Langflow in Google and open it. You need to download Langflow in the desktop version. I have already downloaded Langflow on my laptop, and it looks like this.

As you can see, I have already made one simple agent. First, I will walk you through how I made my simple agent using an offline agent. You can see the agent box here. In this agent box, I am using the language model Llama 3.2.

If I click here and manage model providers, along with Ollama there are other available LLM providers such as Anthropic, Google, OpenAI, and Watson. You can connect any of them using their APIs. Why have I chosen Ollama here? Because Ollama runs on your system and does not give API access or data access to outside systems.

To use Ollama, you also need to install Ollama on your system. Ollama can be installed by downloading and installing the software. This is also covered in one of the previous demos.

As you can see, there are three models in Ollama on my system. These are Llama 3.1 Billion, Llama 3.2 Latest, and Llama 3.8 Billion. Since I am working on my laptop, I have purposefully chosen very small

models. If you are working on an enterprise system, you can easily choose open models such as 70-billion or 240-billion parameter models.

So, for demonstration purposes, I have chosen these three models. Out of these three models, my agent uses one of them. Let me choose Qwen. Again, we have system instructions.

What system instructions are given? 'You are a helpful AI assistant that is used to answer questions and perform tasks.' I am not giving a specific agent here, but you can edit this content and context to create your own agent for business purposes.

As you can see, this is my agent, and where are the inputs coming from? One input is coming from the chat input, which I will soon show you. Another input is coming from the calculator. Another input is coming from the URL fetcher.

You can see there are green dots in the toolsets and a blue dot in the input. The green dots indicate that the calculator and URL fetcher are tools for the agent. You can also see that the response goes into the chat output.

So this is my very simple agent, where I am using a calculator as a tool, DuckDuckGo Search as a tool, and my Qwen 3 8-billion model as the LLM. Just to remind you, you can make this workflow much more complex based on your system. You can add more agents, more system prompts, and so on.

Once you create it, you just click on the playground. I will show you the prompts I gave earlier. I asked, 'Hello, how are you?' The agent responded with the date and time.

Then I asked, 'What is three plus three?' It was not able to respond. Then I changed the model and started asking questions again. I asked, 'Can you tell me what is four into five?' The better model answered that it is 20.

Then I asked this model, 'What is the latest AI research news?' It fetched content from websites using DuckDuckGo Search and provided an answer. Here I can ask any query and run it live.

For example, I asked, 'What is the date today and top headlines of yesterday?' Since it is working on my laptop, even an 8-billion parameter model will take much more time. I am also asking DuckDuckGo Search to search for the information, so it will take some time.

As you can see, we have waited a long time for the output, and it is still running. It has called two tools – getting the current date and fetching the data. It set the current date as 13 May 2026.

For yesterday's top headlines, it is saying that there is breaking news from Fox News and local events. However, the specific headlines are not clearly listed in the retrieved data. It suggests checking the mentioned sources directly and asks whether further clarification is needed.

So you can see the limitations of small agents and small tools. However, this can be mitigated if you use enterprise-grade systems. Your inference can also be much faster if you deploy on GPUs or faster servers.

Basically, you can make your own agents, build complicated agents, and create complicated workflows. To make or add them, you can see here that you can add your agents, embedding models, language models, message history, and LLM operations.

All of this source code can also be pushed to your website, GitHub, or your data repository. Once you have your own downloadable agents from Hugging Face or GitHub, you can use those larger and more sophisticated LLMs to make your workflows much easier.

One can go through these options and try these workflows on simple agents within Langflow.

As we have seen in this session, Langflow can be downloaded directly onto your system. It creates a data-security and sovereignty model because the models are used offline. This is a different way of using LLMs, unlike the more commonly known approach of using only web interfaces.

Based on your own workflow and use cases, you can make your own agents. The underlying concept is that an AI agent perceives a trigger. Maybe it is coming from a user. Maybe it is coming from an event. It reasons using the LLM, then it acts, calls tools, and acts again.

There is also a loop of learning where it learns from your actions and previous data. Using only one LLM is not a limitation. You can add multiple LLMs and give them different system prompts to perform specific jobs.

You can create many agents, give them different personas through system prompts, and automate complicated workflows in this manner.

So basically, both Gumloop and Langflow can be used as sandbox mechanisms to test your automation requirements. Once validated, you can then scale these automation requirements into enterprise environments and beyond.

Video 10: N8N

Hello learners, welcome to this session on a walkthrough of different platforms.

In the previous two sessions, we have gone into the depth of how Gumloop and Langflow work. To recap, in Gumloop we created a very simple agent in a no-code setup. Then, in Langflow, we went through how we can deploy this agent and create it from an offline setup where outside API calls are not required.

Now, with this agentic AI boom, we have several platforms that can do similar kinds of things. Based on the company direction, these platforms provide different kinds of capabilities. Some platforms are enterprise-oriented, so they focus on scalability. Some platforms are user-oriented, so they are good at tool integration, system calling, sandboxing, and so on.

Multiple platforms are emerging on a weekly basis. We will go through some popular platforms.

One of the popular platforms here is N8N. N8N has become very popular over the last six to eight months, and it is one of the leading agentic AI systems. As you have seen in Gumloop and Langflow, its interface is similar.

You can see there is an editor, execution, and evaluation section. We can add different components and AI agents to it. Within AI agents, you can add different LLMs, chat models, and so on. Similar to Gumloop and Langflow, its dashboard looks like this.

You can create a new workflow. I have created two sample workflows here. You can create workflows, chat directly, and access an admin panel, templates, insights, and so on.

Let me go to a created workflow, a sandbox workflow like this one. Like Langflow and Gumloop, it gives you a sandbox environment to try different inputs, outputs, agents, and so on. Then you can publish these as agents, similar to the other agentic AI platforms.

Along with N8N, there are various other platforms. I have chosen Dify as another example. It is very useful when you want to create systems at scale. It is a little more complicated than Gumloop and Langflow.

Here, the orchestration concepts are similar. You can create agents by adding workflows, triggers, and nodes from here. You can also add code, templates, document extractors, logic, and so on.

Similarly, there is another platform, Make.com. This works in a similar way, but it is a little more visual in nature. In one of the previous demos, I showed OpenRouter integrated with Google Sheets. Here, you can add module upon module. There are different toolsets available.

How you open data, how you process it, and how you store the output can all be configured within the Make.com platform. You can build your scenarios and create your agents.

Basically, you can try all these different platforms to create workflow automation. See how they work. If they are working well, you may want to fine-tune them, select the optimal model appropriately, and make choices about the model based on cost, inference time, number of users, enterprise integration capabilities, guardrails, data security, and the ability to address data security concerns.

In short, the AI agent platforms that we have seen in this video and the previous videos have one common theme emerging. They use LLMs to integrate intelligence into your workflows.